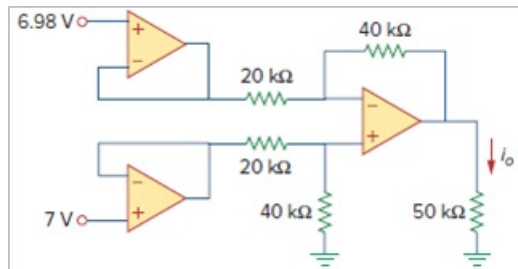


Problem

Obtain i_o in the instrumentation amplifier circuit of Fig. 5.27.

Figure. 5.27: Instrumentation amplifier;



Step-by-step solution

Step 1 of 3

Refer to Figure 5.27 in the text book.

Label the circuit as shown in Figure 1.

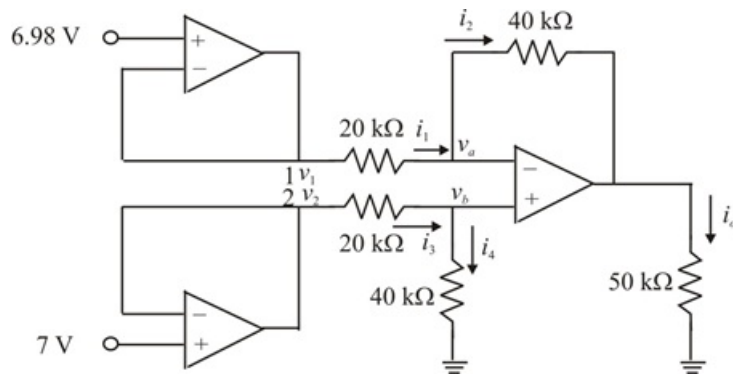


Figure 1

Step 2 of 3

The first 2 op-amp are source followers. Hence the voltage at node 1 is 6.98 V and the voltage at node 2 is 7V. Then the circuit becomes a difference amplifier.

Applying Kirchhoff's current law to node 1,

$$i_1 = i_2$$

$$\frac{v_1 - v_a}{20} = \frac{v_a - v_o}{40}$$

$$2v_1 - 2v_a = v_a - v_o$$

$$v_o = 3v_a - 2v_1$$

$$\text{Therefore } v_o = 3v_a - 2v_1 \dots\dots (1)$$

Applying Kirchhoff's current law to node 2,

$$i_3 = i_4$$

$$\frac{v_2 - v_b}{20} = \frac{v_b - 0}{40}$$

$$2v_2 - 2v_b = v_b$$

$$v_b = 0.66v_2$$

$$\text{Therefore } v_b = 0.66v_2 \dots\dots (2)$$

Step 3 of 3

But $v_a = v_b$ for an ideal op amp, substituting the equation (2) in equation (1), we get

$$\begin{aligned} v_o &= 3v_a - 2v_1 \\ &= 3(0.66v_2) - 2v_1 \\ &= 2(v_2 - v_1) \\ &= 2(7 - 6.98) \\ &= 0.04 \text{ V} \end{aligned}$$

Applying Ohm's law, the current i_o can be calculated as follows:

$$\begin{aligned} i_o &= \frac{v_o}{R} \\ &= \frac{0.04}{50 \times 10^3} \\ &= 800 \text{ nA} \end{aligned}$$

Therefore the current i_o is 800 nA.